

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming

Course Name	Python Programming
Course Code	CSA0804
Name	Monish.L
Register Number	1972260035B
Topic	CO5-AT2-DEBUGGING AND OPTIMIZATION TASK
Question	Q2 — Corrupted DataFrame Cleaning Pipeline (5 Marks) — Pandas – Missing Values, Duplicates

Q2. Corrupted DataFrame Cleaning Pipeline [5 Marks] — Pandas – Missing Values, Duplicates

■ This program contains 5 bug(s). Identify, fix, and optimise.

- (a) Identify ALL bugs and logical errors in the pipeline.
- (b) Fix and optimise: use `fillna()`, `drop_duplicates()`, and `reset_index()` correctly.
- (c) Print cleaned shape vs original shape.

Run the Program:

```
import pandas as pd
df = pd.read_csv('employees.csv')
# Remove duplicates
df.drop_duplicates(inplace=True)
# Fill missing salary with mean
mean_sal = df['Salary'].means()
df['Salary'].fillna(mean_sal)
# Fill missing department with 'Unknown'
df['Department'] = df['Department'].fillna()
# Reset index
df.reset_index(drop=False, inplace=True)
print('Cleaned rows:', df.shape[1])
print(df.isnull().sum())
```

(a) Bugs and Logical Errors Identified

Bug 1 – Wrong method name (Salary mean):

`df['Salary'].means()` calls a method that does not exist on a pandas Series (**AttributeError**). The correct method is `.mean()`, no 's'.

Bug 2 – `fillna()` result discarded (Salary):

`df['Salary'].fillna(mean_sal)` returns a new Series but it is never assigned back and **inplace** is not set, so the original **Salary** column still has its missing values. It must be reassigned: `df['Salary'] = df['Salary'].fillna(mean_sal)`.

Bug 3 – `fillna()` missing fill value (Department):

`df['Department'].fillna()` is called with no argument, which raises a **TypeError** (fillna needs a value to fill with). Per the comment above it, it should be `df['Department'].fillna('Unknown')`.

Bug 4 – reset_index() keeps the old index (logical error):

`df.reset_index(drop=False, inplace=True)` uses `drop=False`, which inserts the old (pre-duplicate-removal) index back in as a new 'index' column instead of discarding it. To cleanly reset the index it should be `drop=True`.

Bug 5 – Wrong shape axis for row count (logical error):

`print('Cleaned rows:', df.shape[1])` prints the number of **columns**, not rows — `df.shape` is (**rows**, **columns**), so the row count is `df.shape[0]`, not `df.shape[1]`.

(b) Corrected and Optimised Program

The program below fixes all five bugs, correctly fills and reassigns missing values, resets the index cleanly, and reports the cleaned shape against the original shape as required by part (c).

```
"""
C05-AT2 - Debugging and Optimisation Task
Challenge 2: Corrupted DataFrame Cleaning Pipeline
[Pandas - Missing Values, Duplicates]
-----

Loads employee data from CSV, removes duplicate rows, fills
missing Salary and Department values, resets the index, and
reports the cleaned shape against the original shape.
"""

import pandas as pd

# -----
# 1. Load the CSV file and record the original shape
# -----
df = pd.read_csv('employees.csv')
original_shape = df.shape

# -----
# 2. Remove duplicate rows
# -----
df.drop_duplicates(inplace=True)

# -----
# 3. Fill missing salary with the column mean
#     (fixed: .mean() not .means(), result reassigned)
# -----
mean_sal = df['Salary'].mean()
df['Salary'] = df['Salary'].fillna(mean_sal)

# -----
# 4. Fill missing department with 'Unknown'
#     (fixed: fill value supplied)
# -----
df['Department'] = df['Department'].fillna('Unknown')

# -----
# 5. Reset index after dropping duplicates
#     (fixed: drop=True discards the old index instead of
```

```

#     inserting it as a new column)
# -----
df.reset_index(drop=True, inplace=True)

# -----
# 6. Report cleaned shape vs original shape (part c)
#     (fixed: shape[0] is the row count, not shape[1])
# -----
print('Original shape:', original_shape)
print('Cleaned shape:', df.shape)
print('Cleaned rows:', df.shape[0])

print('\nRemaining null values per column:')
print(df.isnull().sum())

```

(c) Cleaned vs Original Shape

original_shape is captured immediately after loading, before any cleaning, so it reflects the raw row/column count.

df.shape after cleaning shows the effect of dropping duplicates (fewer rows, same columns).

df.shape[0] (not **shape[1]**) is printed as the row count, since **shape** is **(rows, columns)**.

df.isnull().sum() confirms Salary and Department no longer contain missing values after the `fillna()` fixes.

Expected Sample Output

Exact values depend on the contents of **employees.csv**. For a file with columns *Name*, *Department*, *Salary*, a few duplicate rows, and some missing Salary/Department entries, the output would look like:

```

Original shape: (15, 3)
Cleaned shape: (12, 3)
Cleaned rows: 12

Remaining null values per column:
Name          0
Department    0
Salary        0
dtype: int64

```

Conclusion

The original pipeline looked reasonable at a glance but failed or misbehaved at every stage: a misspelled **.means()** call would crash before the mean could even be computed, the Salary `fillna()` result was thrown away instead of reassigned, the Department `fillna()` was missing the very fill value its own comment described, **reset_index()** kept the stale index as a stray column, and the final print read the column count instead of the row count. The corrected version reassigns every `fillna()` result, supplies the missing fill value, resets the index with **drop=True**, and reports **shape[0]** so the cleaned row count and original shape can be compared accurately.